

Variable Selection and Model Building

Model-Building Problem

The best strategy for building a regression model is:

1. Fit the full model (the model with all of the regressors under consideration)
2. Perform a thorough analysis of this model, including a full residual analysis.
3. Determine if transformations of the response or of some of the regressors are necessary.
4. Use the t tests on the individual regressors to edit the model.
5. Perform a thorough analysis of the edited model, especially a residual analysis, to determine the model's adequacy.

In most practical problems, especially those involving historical data, the analyst has a rather large pool of possible candidate regressors, of which only a few are likely to be important.

Finding an appropriate subset of regressors for the model is often called the **variable selection problem**

Criteria for Evaluating Subset Regression Models

In this section we will discuss the criteria for selecting and comparing subset regression models.

Coefficient of Multiple Determination

A measure of the adequacy of a regression model that has been widely used is the coefficient of multiple determination, R^2

Let R_p^2 denote the coefficient of multiple determination for a subset regression model with p terms that is $p - 1$ regressors and an intercept β_0 .

Computationally

$$R_p^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Models that have large values of R_p^2 are preferred.

Adjusted R^2

To avoid the difficulties of interpreting R^2 some analysts prefer to use the adjusted R^2 statistic defined by:

$$R_\alpha^2 = 1 - \left(\frac{n-1}{n-p} \right) (1 - R_p^2)$$

Models that have large values of R_α^2 are preferred.

Residual Mean Square

The residual mean square for a subset regression model is given by

$$MS_{Res} = \frac{SS_{Res}}{n - p}$$

may be used in model selection. The model with the smallest MS_{Res} are preferred.

Akaike Information Criterion (AIC)

The AIC is a penalized log-likelihood measure. Let L be the likelihood function for a specific model. The AIC is defined as:

$$AIC = -2 \ln(L) + 2p$$

where p is the number of parameters in the model.

The smaller the AIC the better the model

Bayesian Information Criterion (BIC)

This criterion places a greater penalty on adding regressors as the sample size increases.

The criterion is:

$$BIC = -2 \ln L + p \ln(n)$$

The smaller the BIC the better the model

Variable Selection Criteria

Variable selection means selecting which variables to include in our model.

We explore the variable selection criteria in details:

All Possible Regressions

For all candidate regressors fit all possible models.

If there are K candidate regressors fit and examine $2^K - 1$ models and choose the best model based on the discussed criteria of model selection

Example

The `mtcars` data in R gives the fuel consumption of 32 automobiles in US. We are to fit the multiple linear regression model using the variables:

- Y_{mpg} gives Miles per US gallon
- $X_1 \text{ displ}$ displacement
- $X_2 \text{ hp}$ gross horsepower

Since we have 2 predictors we are to fit $2^2 - 1 = 3$ models given as:

i $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

ii $Y = \beta_0 + \beta_1 X_1 + \varepsilon$

iii $Y = \beta_0 + \beta_1 X_2 + \varepsilon$

We fit the models and extract the AIC, MSE, R^2 and R_α^2

```
# fitting the model

model1 <- lm(mpg ~ disp + hp , data = mtcars)

model2 <- lm(mpg ~ disp , data = mtcars)

model3 <- lm(mpg ~ hp , data = mtcars)
```

Model	AIC	MSE	R^2	R_α^2
X_1, X_2	168.16	3.127	0.7482	0.7309
X_1	170.20	3.251	0.7183	0.709
X_2	181.23	3.386	0.6024	0.589

Selection using AIC, MSE, R^2 and R_α^2 all yield the same results so the optimal model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Alternatively can do:

```
model <- lm(mpg ~ disp + hp, data = mtcars)
ols_step_best_subset(model)
```

Stepwise Regression Methods

Evaluating all possible regressions can be burdensome computationally and for the analyst.

An alternative might be to compare a scientifically meaningful subset of models. The three categories are:

- Forward Selection
- Backward Elimination
- Stepwise Regression

Forward Selection Begin with the assumption that there are no regressors in the model.

Check models with all possible regressors added individually.

Add the regressor that most changes your criterion in the correct direction. Go back to 2.

If none of the regressors have a positive effect on your criterion, stop with the regressors you have. This is your final model.

In R we use the following code:

```
model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
# stepwise forward regression
ols_step_forward_p(model)
```

Backward Selection Start with all p candidate regressors in the model

Drop the predictor that improves your selection criterion the least

Continue until there is no predictor that can be dropped and result in an improvement of your selection criterion, then all the remaining predictors define your final model.

```
model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
# stepwise backward regression
ols_step_backward_p(model)
```

Stepwise Regression General stepwise regression techniques are usually a combination of backward elimination and forward selection, alternating between the two techniques at different steps.

In R we do as follows:

```
library(MASS)
model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
stepAIC(model)
```